

On the Invariant Theory of Forms in n Variables

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§ 9. Form Rows. Reduction Theorems^[152 : *].

We now define the *form row*¹ introduced at the end of § 6 somewhat more precisely as “the totality of *invariant* formations, linear in the coefficients, of a given initial form; that is, the totality of forms obtained from the initial form by folding it in itself, *in the ordering by higher forms*.”

To explain the higher forms, let the initial form, by Theorem VII, be given as the product of two normal forms $S \cdot T$. By a *higher form* we shall mean a form with a higher folding in itself, or, among forms with the same folding in themselves, one specially normalized. More precisely, suppose that the form A has arisen through α_ρ fundamental foldings of type ρ , and the form B through β_ρ such foldings. Then B is higher than A :

- 1) if, in the coordinatewise system of inequalities $\beta_\rho \geq \alpha_\rho$, $\rho = 1, 2, \dots, n-1$, strict inequality holds for at least one value of ρ ^[152 : **];
- 2) if equality holds for every value of ρ , but *fewer* symbol rows have been combined into a single matrix product. This normalization is necessary because of the reductions in § 8. For example, $(S^{n-1} \mid T_{n-1})$ is higher than $(S^{n-1} \mid [S^1 T^\rho]_{n-\rho-1 p_\rho})$.
- 3) if equality holds for every value of ρ and the number of symbol rows combined into a single matrix product is the same, then B is ranked higher than A by a normalization that is arbitrary in itself but fixed once and for all. Such a normalization is always possible because only finitely many forms belong to a value system α ; it may, for example, be achieved by giving priority to one form S (a larger number of symbol rows S , a larger sum of the upper weight indices of the rows S , and so on).

It is useful to picture the form row as an $(n-1)$ -dimensional manifold of lattice points, with the $n-1$ directions corresponding to the $n-1$ fundamental foldings. Each form is then assigned one and only one value system α , that is, a positive integral lattice point.² The ordering of the different forms belonging to a value system α is uniquely fixed by the normalization above. Any form in the form row is the initial form of a new form row contained as a part of the total form row: namely, the part consisting of the forms obtained by proceeding from the new initial form in the $n-1$ positive directions.

By Theorem VIII and the general theory of series expansions, precisely the reduction theorems of the binary and ternary domains (cf. dissertation, § 3) now hold for form rows. We shall briefly derive the principal theorem, first giving the definition:

“In a given system of forms, let some finite number of forms be combined into a module. We call a form *reducible* if it can be expressed in terms of forms having invariants as factors or in terms

^[152 : *]Cf. the corresponding considerations in the ternary domain, dissertation, §§ 1–3.

¹**R823 source defects N04-S09-SRCDEF-001–012:** R823 has six emphasis-defect classes comprising nine printed runs at lines 4478, 4480, 4483, 4490, 4492, and 4494; it also loses two punctuation marks and a page-local note identity, moves two note calls past their printed punctuation, and replaces the printed list labels 1), 2), 3) by 1., 2., 3. The English follows the print; the German authority remains unchanged.

^[152 : **]If some of the comparisons have the sign $>$ and others the sign $<$, the forms are not comparable, since a form obtained from another by folding always has a positive value system of fundamental foldings; hence, in our case, the values $\beta_\rho - \alpha_\rho$ are positive or zero.

²**Source ambiguity N04-S09-AMB-001:** “positive integral” is retained literally, although the initial form has the zero value system. The source does not resolve whether zero coordinates are included under this wording.

of ‘higher forms’; that is, higher forms in the total form row to which the form belongs, or forms containing the symbols of the module to a higher order. We call a reducible form row a *reducer*.³ Then:

Theorem IX: If the initial form of a form row is reducible because it has arisen by folding with a reducer, then the entire form row arising from that initial form is reducible.

For the proof, observe that a form h obtained by folding a form f with a second form g may, by (45), be regarded as a form f containing several cogredient variable rows p_ρ, p'_ρ .³ By the general theory of series expansions, such forms can be represented as polars of the elementary covariants of f , by applying the series expansion successively to each pair of cogredient variable rows p_ρ, p'_ρ . By (38), the elementary covariants that occur are the forms generated by cogredient folding. Since the order in which the series expansion is applied remains arbitrary, we may in particular choose an order corresponding to the generation of general foldings from fundamental foldings. The resulting system of elementary covariants is then identical with the form row f ; the forms generated by cogredient folding of higher defect take their place among the forms generated by fundamental foldings.

Polars of the form row f are, however, also obtained with any order of series expansion, to which a second system of elementary covariants may correspond. Since the form row exhausts the totality of invariant formations, every form in the second system can be expressed linearly in terms of polars of the form row, specifically polars of those forms that exhibit the same or a greater weight decrease. This last assertion follows because (cf. § 7) the polars may be regarded as simultaneous invariants of a form row H in (52), with degrees corresponding to the form h , and of the form row f . Every folding of H in itself corresponds to a weight decrease that, after substitution (11) is performed, is transferred to the symbol rows and hence to the forms of f ^[154 : *].

An arbitrary form of the form row h has arisen by folding h in itself; that is, by a higher folding of g with f on the one hand, and by folding f or g in itself on the other. In both cases the series expansion leads to polars of the form row f , just as shown above for the initial form h . In particular, if f is a reducer, an expansion in polars of the reducer results; hence the form row h is reducible.

Applications of the theorem to the reduction of complete systems of forms follow analogously to the binary and ternary domains.

³**Source ambiguity N04-S09-AMB-002:** the print and R823 line 4494 reuse f while describing h . The target preserves that symbol rather than silently renaming it.

^[154 : *]This consideration also underlies the second formulation of the concept of equivalence at the end of § 8. Further discussion of the different systems of elementary covariants in the ternary domain is found in *Study*, loc. cit., II, § 7, Theorems 7) and 8). By our Theorem VII, the same results also hold for n variables.